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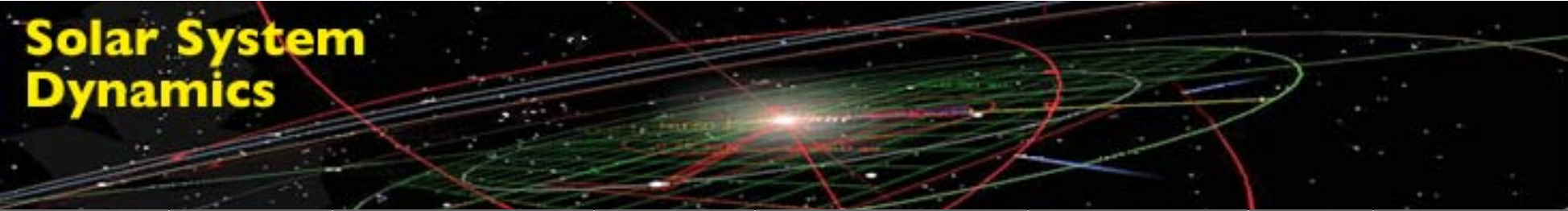
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Planetary Satellite Mean Orbital Parameters
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WARNING: These mean orbital parameters are **not intended for ephemeris computation**. Accurate ephemerides should be obtained from our [HORIZONS](#) system. Mean orbital parameters are primarily useful in describing the general shape and orientation of a planetary satellite's orbit.

Satellites of Earth

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Mean [ecliptic](#) orbital elements

Epoch 2000 Jan. 1.50 TT

Solution: DE405/LE405

Sat.	<u>a</u>	<u>e</u>	<u>w</u>	<u>M</u>	<u>i</u>	<u>node</u>	<u>n</u>	<u>P</u>	<u>P_w</u>	<u>P_{node}</u>	Ref.
	(km)		(deg)	(deg)	(deg)	(deg)	(deg/day)	(days)	(yr)	(yr)	
Moon	384400.	0.0554	318.15	135.27	5.16	125.08	13.176358	27.322	5.997	18.600	1

Satellites of Mars

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Mean orbital elements referred to the local [Laplace planes](#)

Epoch 1950 Jan. 1.00 TT

Solution: MAR080

Sat.	<u>a</u>	<u>e</u>	<u>w</u>	<u>M</u>	<u>i</u>	<u>node</u>	<u>n</u>	<u>P</u>	<u>P_w</u>	<u>P_{node}</u>	<u>R.A.</u>	<u>Dec.</u>	<u>Tilt</u>	Ref.
	(km)		(deg)	(deg)	(deg)	(deg)	(deg/day)	(days)	(yr)	(yr)	(deg)	(deg)	(deg)	
Phobos	9376.	0.0151	150.057	91.059	1.075	207.784	1128.8447569	0.3189	1.1316	2.2617	317.671	52.893	0.009	5
Deimos	23458.	0.0002	260.729	325.329	1.788	24.525	285.1618790	1.2624	27.3703	54.5367	316.657	53.529	0.889	5

Satellites of Jupiter

jump to: [Earth](#), [Mars](#), [Jupiter](#), [Saturn](#), [Uranus](#), [Neptune](#), [Pluto](#) or [\[top of page\]](#)

Mean orbital elements referred to the local [Laplace planes](#)

Epoch 1997 Jan. 16.00 TT

Sat.	<u>a</u>	<u>e</u>	<u>w</u>	<u>M</u>	<u>i</u>	<u>node</u>	<u>n</u>	<u>P</u>	<u>P_w</u>	<u>P_{node}</u>	<u>R.A.</u>	<u>Dec.</u>	<u>Tilt</u>	Ref.
	(km)		(deg)	(deg)	(deg)	(deg)	(deg/day)	(days)	(yr)	(yr)	(deg)	(deg)	(deg)	
Io	421800.	0.0041	84.129	342.021	0.036	43.977	203.4889583	1.769	1.625	7.420	268.057	64.495	0.000	11

Europa	671100.	0.0094	88.970	171.016	0.466	219.106	101.3747242	3.551	1.394	30.184	268.084	64.506	0.016	11
Ganymede	1070400.	0.0013	192.417	317.540	0.177	63.552	50.3176072	7.155	63.549	132.654	268.168	64.543	0.068	11
Callisto	1882700.	0.0074	52.643	181.408	0.192	298.848	21.5710728	16.69	205.75	338.82	268.639	64.749	0.356	11
Amalthea	181400.	0.0032	155.873	185.194	0.380	108.946	722.6317143	0.498	0.196	0.393	268.057	64.495	0.000	11
Thebe	221900.	0.0176	234.269	135.956	1.080	235.694	533.7002568	0.675	0.398	0.797	268.057	64.495	0.000	11
Adrastea	129000.	0.0018	328.047	135.673	0.054	228.378	1206.9988064	0.298	0.058	0.116	268.057	64.496	0.000	11
Metis	128000.	0.0012	297.177	276.047	0.019	146.912	1221.2545982	0.295	0.057	0.115	268.057	64.496	0.000	11

[Ecliptic](#) mean orbital elements

Epoch 2000 Jan. 1.00 TT
Solution: [Ecliptic](#) mean elements - JUP300

Sat.	a	e	w	M	i	node	n	P	P_w	P_{node}	Ref.
	(<i>km</i>)		(<i>deg</i>)	(<i>deg</i>)	(<i>deg</i>)	(<i>deg</i>)	(<i>deg/day</i>)	(<i>days</i>)	(<i>yr</i>)	(<i>yr</i>)	
Himalia	11460000.	0.1586	331.234	66.874	28.612	64.798	1.4367562	250.56	139.38	292.57	18
Elara	11740000.	0.2108	142.001	330.985	27.945	115.511	1.3865315	259.64	128.07	265.30	18
Pasiphae	23629000.	0.4062	169.226	279.769	151.413	314.223	0.4841223	743.61	81.20	81.17	18
Sinope	23942000.	0.2552	354.541	165.352	158.189	309.199	0.4743772	758.89	88.72	87.40	18
Lysithea	11717000.	0.1161	49.670	330.475	27.663	5.326	1.3888891	259.20	131.19	292.04	18
Carme	23401000.	0.2546	26.416	233.375	164.994	114.854	0.4903501	734.17	80.98	88.32	18
Ananke	21254000.	0.2332	95.772	253.384	148.693	15.959	0.5716125	629.80	144.92	110.12	18
Leda	11164000.	0.1624	269.393	230.352	27.882	219.181	1.4942407	240.93	144.79	301.94	18
Callirrhoe	24099000.	0.2796	23.909	107.962	147.080	283.104	0.4744232	758.82	120.06	89.03	18
Themisto	7504000.	0.2435	217.147	313.051	42.977	192.288	2.7688033	130.02	686.41	528.39	18
Megaclite	23814000.	0.4156	288.882	135.272	152.781	280.575	0.4781658	752.88	76.90	79.23	18
Taygete	23363000.	0.2523	231.540	94.756	165.253	305.114	0.4915209	732.42	81.04	88.59	18
Chaldene	23181000.	0.2503	243.878	267.454	165.155	134.240	0.4974208	723.73	81.66	89.52	18
Harpalyke	21106000.	0.2296	134.505	215.956	148.759	29.834	0.5775515	623.32	147.17	111.64	18
Kalyke	23565000.	0.2466	218.934	255.702	165.121	43.864	0.4851490	742.04	80.76	87.67	18
Iocaste	21272000.	0.2152	64.727	213.675	149.411	269.613	0.5699784	631.60	145.78	111.18	18
Erinome	23286000.	0.2655	10.274	267.136	164.914	317.497	0.4941719	728.49	80.51	88.33	18
Isonoe	23231000.	0.2471	116.879	124.941	165.250	130.961	0.4956884	726.26	81.80	89.52	18
Praxidike	21148000.	0.2274	190.862	117.480	148.885	280.956	0.5756381	625.39	144.77	111.16	18
Autonoe	24037000.	0.3152	54.793	142.035	152.364	272.817	0.4730563	761.01	92.33	85.00	18
Thyone	21197000.	0.2307	97.023	238.786	148.595	233.022	0.5739872	627.19	147.58	110.93	18
Hermippe	21297000.	0.2095	300.836	131.854	150.740	330.393	0.5679040	633.91	136.92	110.64	18
Aitne	23317000.	0.2627	99.401	105.000	165.048	8.679	0.4930702	730.12	80.36	88.19	18
Eurydome	23146000.	0.2755	223.631	287.689	150.271	302.470	0.5018735	717.31	111.58	93.30	18
Euanthe	21039000.	0.2320	320.635	333.101	148.915	254.297	0.5802269	620.45	144.52	111.71	18
Euporie	19336000.	0.1438	89.904	70.243	145.740	60.143	0.6537221	550.69	337063.88	139.45	18
Orthosie	21158000.	0.2807	216.805	204.517	146.004	221.949	0.5782402	622.58	142.35	107.53	18
Sponde	23790000.	0.3112	61.885	174.044	150.997	116.363	0.4810760	748.32	98.08	87.01	18
Kale	23306000.	0.2597	44.233	212.853	164.944	60.170	0.4934152	729.61	80.91	88.57	18
Pasithee	23091000.	0.2682	231.920	215.443	165.117	327.729	0.5003695	719.47	80.82	89.25	18
Hegemone	23575000.	0.3445	197.144	236.950	154.164	318.902	0.4866038	739.82	85.66	84.87	18
Mneme	21033000.	0.2258	40.542	256.860	148.585	13.467	0.5806001	620.05	148.06	112.26	18
Aoede	23974000.	0.4325	59.739	197.676	158.272	173.392	0.4728089	761.41	69.17	76.83	18
Thelxinoe	21160000.	0.2201	313.183	268.013	151.390	169.962	0.5732248	628.03	130.58	110.42	18

Arche	23352000.	0.2495	171.632	39.713	165.015	339.210	0.4918707	731.90	81.40	88.73	18
Kallichore	23276000.	0.2509	9.836	55.937	165.102	30.339	0.4943462	728.23	81.56	89.16	18
Helike	21065000.	0.1498	299.482	43.659	154.842	89.749	0.5747808	626.33	127.40	114.52	18
Carpo	17056000.	0.4317	90.372	337.062	51.624	50.597	0.7889862	456.28	711550.25	112.82	18
Eukelade	23323000.	0.2619	309.685	204.846	165.265	193.558	0.4929260	730.33	80.24	88.16	18
Cyllene	23800000.	0.4155	187.429	128.345	150.336	252.611	0.4787365	751.98	80.19	79.71	18
Kore	24482000.	0.3313	138.071	33.416	145.173	313.355	0.4634134	776.84	108.39	84.47	18
Herse	23408000.	0.2541	330.295	141.667	164.964	295.702	0.4901154	734.52	80.98	88.27	18
S/2000 J 11	12297000.	0.2320	173.544	309.734	28.631	294.497	1.2939791	278.21	117.58	241.93	18
S/2003 J 2	28347000.	0.4100	165.201	237.932	157.291	344.782	0.3671484	980.53	60.25	62.87	18
S/2003 J 3	20221000.	0.1969	66.338	311.780	147.547	231.489	0.6165779	583.87	181.96	122.74	18
S/2003 J 4	23929000.	0.3624	197.401	260.480	149.589	179.131	0.4766663	755.25	90.17	82.96	18
S/2003 J 5	23495000.	0.2476	90.066	336.636	165.248	176.683	0.4873118	738.75	80.95	88.10	18
S/2003 J 9	23385000.	0.2632	292.662	348.415	165.047	44.321	0.4909185	733.32	80.27	87.93	18
S/2003 J 10	23042000.	0.4299	170.833	258.937	165.073	151.911	0.5026172	716.25	68.06	80.34	18
S/2003 J 12	17830000.	0.4904	13.288	38.543	151.003	65.530	0.7351860	489.67	97.93	109.42	18
S/2003 J 15	22627000.	0.1899	18.405	58.865	146.492	236.674	0.5219082	689.78	187.83	105.37	18
S/2003 J 16	21097000.	0.2281	57.681	307.563	148.683	16.883	0.5779589	622.88	146.70	111.64	18
S/2003 J 18	20508000.	0.0895	130.894	202.160	146.077	158.247	0.6018719	598.13	506.86	131.88	18
S/2003 J 19	23533000.	0.2552	176.668	223.035	165.116	27.442	0.4862156	740.41	80.17	87.39	18
S/2003 J 23	23567000.	0.2746	255.114	144.222	146.424	41.706	0.4914942	732.46	130.10	92.90	18
S/2010 J 1	23449000.	0.2491	189.230	160.525	165.100	282.871	0.4887995	736.50	81.18	88.33	18
S/2010 J 2	21004000.	0.2267	18.252	312.074	148.673	5.802	0.5817371	618.84	147.06	112.35	18
S/2011 J 1	23446000.	0.2534	31.514	256.027	165.318	250.728	0.4888998	736.35	80.45	87.91	18
S/2011 J 2	23124000.	0.3493	270.154	285.597	153.597	24.866	0.5011321	718.37	87.78	87.00	18

Satellites of Saturn

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Mean orbital elements referred to the local [Laplace planes](#)

Epoch 2000 Jan. 1.50 TT

Sat.	<u>a</u>	<u>e</u>	<u>w</u>	<u>M</u>	<u>i</u>	<u>node</u>	<u>n</u>	<u>P</u>	<u>P_w</u>	<u>P_{node}</u>	<u>R.A.</u>	<u>Dec.</u>	<u>Tilt</u>	Ref.
	(<i>km</i>)		(<i>deg</i>)	(<i>deg</i>)	(<i>deg</i>)	(<i>deg</i>)	(<i>deg/day</i>)	(<i>days</i>)	(<i>yr</i>)	(<i>yr</i>)	(<i>deg</i>)	(<i>deg</i>)	(<i>deg</i>)	
Mimas	185539.	0.0196	332.499	14.848	1.574	173.027	381.9944948	0.942	0.493	0.986	40.589	83.536	0.002	32
Enceladus	238042.	0.0000	0.076	199.686	0.003	342.507	262.7318978	1.370	1.184	2.360	40.586	83.536	0.002	32
Tethys	294672.	0.0001	45.202	243.367	1.091	259.842	190.6979109	1.888	2.490	4.982	40.578	83.537	0.001	32
Dione	377415.	0.0022	284.315	322.232	0.028	290.415	131.5349307	2.737	5.852	11.709	40.544	83.540	0.005	32
Rhea	527068.	0.0002	241.619	179.781	0.333	351.042	79.6900459	4.518	17.939	35.832	40.328	83.559	0.036	32
Titan	1221865.	0.0288	180.532	163.310	0.306	28.060	22.5769756	15.95	352.12	704.60	36.214	83.949	0.629	32
Hyperion	1500933.	0.0232	303.178	86.342	0.615	263.847	16.9199503	21.28	127.99	154.66	36.372	83.862	0.564	32
Iapetus	3560854.	0.0293	271.606	201.789	8.298	81.105	4.5379416	79.33	1676.69	3438.73	284.715	78.749	15.210	32
Phoebe	12947918.	0.1634	342.500	53.038	175.243	241.086	0.6569114	548.02	460.92	723.17	275.954	67.455	26.723	32
Janus	151450.	0.0098	16.012	17.342	0.165	154.175	518.3431513	0.695	0.240	0.482	40.585	83.538	0.000	33
Epimetheus	151450.	0.0161	88.975	80.377	0.353	192.762	518.0979676	0.695	0.240	0.482	40.585	83.538	0.000	33
Helene	377444.	0.0000	33.134	43.186	0.213	163.112	131.5349306	2.737	2.562	11.706	40.544	83.540	0.005	32
Telesto	294720.	0.0002	119.135	260.157	1.180	229.182	190.6979089	1.888	2.489	4.982	40.578	83.537	0.001	32
Calypso	294721.	0.0005	17.470	156.660	1.500	314.226	190.6979089	1.888	2.487	4.983	40.578	83.537	0.001	32
Atlas	137774.	0.0011	210.851	283.282	0.003	236.422	598.3147505	0.602	0.171	0.344	40.585	83.538	0.000	33

Prometheus	139429.	0.0022	37.514	96.886	0.007	319.176	587.2860142	0.613	0.179	0.359	40.585	83.538	0.000	33
Pandora	141810.	0.0042	66.248	125.112	0.050	147.272	572.7878754	0.629	0.190	0.381	40.585	83.538	0.000	33
Pan	133585.	0.0000	103.331	351.187	0.000	52.076	626.0317341	0.575	0.154	0.309	40.583	83.538	0.000	33
Methone	194402.	0.0000	134.636	71.189	0.013	313.562	356.5949252	1.010	0.250	1.158	40.589	83.536	0.002	32
Pallene	212282.	0.0040	16.074	356.229	0.001	123.180	312.0270737	1.154	0.862	1.899	40.579	83.537	0.001	33
Polydeuces	377222.	0.0191	311.847	89.307	0.175	67.936	131.5349306	2.737	5.811	11.695	40.543	83.540	0.005	33
Daphnis	136504.	0.0000	266.931	113.790	0.003	132.867	605.9791512	0.594	0.166	0.334	40.583	83.538	0.000	33
Anthe	196888.	0.0011	138.902	190.473	0.015	287.852	347.3587167	1.036	0.617	1.232	40.583	83.537	0.000	33
Aegaeon	167425.	0.0002	152.905	322.771	0.001	317.202	445.4934177	0.808	0.343	0.688	40.584	83.537	0.001	33

Mean [ecliptic](#) orbital elements

Epoch 2000 Jan. 1.00 TT

Sat.	<u><i>a</i></u>	<u><i>e</i></u>	<u><i>w</i></u>	<u><i>M</i></u>	<u><i>i</i></u>	<u><i>node</i></u>	<u><i>n</i></u>	<u><i>P</i></u>	<u><i>P_w</i></u>	<u><i>P_{node}</i></u>	Ref.
	<i>(km)</i>		<i>(deg)</i>	<i>(deg)</i>	<i>(deg)</i>	<i>(deg)</i>	<i>(deg/day)</i>	<i>(days)</i>	<i>(yr)</i>	<i>(yr)</i>	
Ymir	23128000.	0.3338	21.352	228.673	173.497	192.937	0.2737369	1315.13	217.61	277.53	34
Paaliaq	15204000.	0.3325	237.522	321.654	46.228	330.022	0.5240567	686.95	576.14	535.71	34
Tarvos	18243000.	0.5382	274.104	265.783	33.725	102.504	0.3886226	926.35	194.18	272.97	34
Ijiraq	11408000.	0.2717	92.899	17.328	47.485	130.779	0.7974868	451.42	13807.37	935.12	34
Suttungr	19468000.	0.1139	34.281	321.133	175.815	227.259	0.3540952	1016.68	292.79	402.20	34
Kiviuq	11384000.	0.3325	90.205	172.018	46.764	353.584	0.8013951	449.22	19973.94	840.86	34
Mundilfari	18653000.	0.2097	309.694	92.821	167.439	82.856	0.3778328	952.80	312.23	417.31	34
Albiorix	16393000.	0.4797	55.932	32.828	34.060	102.512	0.4595027	783.46	249.88	358.14	34
Skathi	15635000.	0.2718	203.517	114.689	152.633	286.599	0.4944382	728.10	474.36	546.69	34
Erriapus	17602000.	0.4722	282.522	294.829	34.481	150.985	0.4132539	871.14	219.90	323.49	34
Siarnaq	18182000.	0.2802	65.929	201.288	45.809	47.826	0.4020067	895.51	457.40	454.20	34
Thrymr	20418000.	0.4659	125.404	30.075	177.659	285.762	0.3296678	1092.01	875.11	139002.09	34
Narvi	19349000.	0.4296	169.959	114.172	145.731	174.435	0.3585873	1003.94	341.38	347.42	34
Aegir	20751000.	0.2524	242.651	26.017	166.668	179.064	0.3220526	1117.83	273.91	350.18	34
Bebhionn	17116000.	0.4682	358.141	168.045	35.101	199.128	0.4312114	834.86	236.57	342.06	34
Bergelmir	19336000.	0.1420	133.400	306.494	158.557	202.164	0.3579396	1005.76	359.70	424.95	34
Bestla	20209000.	0.5145	81.185	239.156	145.136	288.308	0.3308757	1088.02	284.99	299.08	34
Farbauti	20390000.	0.2414	342.995	282.813	156.520	135.109	0.3312924	1086.65	330.77	374.34	34
Fenrir	22454000.	0.1347	120.982	131.678	164.963	226.595	0.2856343	1260.35	276.77	332.99	34
Fornjot	25146000.	0.2077	324.787	214.499	170.372	259.946	0.2409497	1494.09	223.68	267.44	34
Hati	19868000.	0.3710	21.286	163.640	165.808	324.380	0.3460954	1040.18	262.48	339.97	34
Hyrrokkin	18440000.	0.3359	273.076	291.841	151.536	45.402	0.3863203	931.87	375.76	406.45	34
Kari	22093000.	0.4756	163.935	286.021	156.067	281.211	0.2924012	1231.18	228.81	270.06	34
Loge	23059000.	0.1862	32.821	337.237	167.689	343.811	0.2745228	1311.37	251.72	307.35	34
Skoll	17668000.	0.4636	193.115	44.965	161.010	296.623	0.4099120	878.24	286.68	369.06	34
Surtur	22941000.	0.4459	303.662	136.191	169.688	236.537	0.2775384	1297.12	202.95	259.66	34
Jarnsaxa	19354000.	0.2178	237.422	198.750	163.649	22.519	0.3575194	1006.94	312.48	396.92	34
Greip	18457000.	0.3146	152.160	314.541	174.800	349.350	0.3841491	937.14	280.23	385.90	34
Tarqeq	17962000.	0.1675	34.767	161.020	46.291	83.291	0.4067094	885.15	896.54	593.15	34
S/2004 S 7	21000000.	0.5290	84.036	79.762	165.693	341.236	0.3157314	1140.21	211.18	273.22	34
S/2004 S 12	19886000.	0.3268	87.128	1.599	165.261	307.942	0.3441326	1046.11	273.22	351.49	34
S/2004 S 13	18406000.	0.2591	346.186	41.077	168.798	205.701	0.3856043	933.60	302.01	409.39	34
S/2004 S 17	19448000.	0.1795	180.792	228.545	168.239	26.664	0.3548169	1014.61	300.99	397.14	34

S/2006 S 1	18780000.	0.1412	154.950	96.596	156.180	336.641	0.3741294	962.23	405.14	453.57	34
S/2006 S 3	22428000.	0.3792	188.728	167.147	158.631	206.993	0.2868177	1255.15	245.54	287.39	34
S/2007 S 2	16718000.	0.1791	57.720	84.066	174.057	111.277	0.4454603	808.15	342.95	496.45	34
S/2007 S 3	18938000.	0.1853	111.854	292.691	177.595	276.824	0.3681825	977.78	1079.63	161352.07	34

Satellites of Uranus

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Mean equatorial orbital elements

Epoch 1980 Jan. 1.0 TT
Solution: GUST86 - Laskar and Jacobson (1987) A&A 188, pp. 212-224 (converted to ICRF)

Sat.	<u>a</u>	<u>e</u>	<u>w</u>	<u>M</u>	<u>i</u>	<u>node</u>	<u>n</u>	<u>P</u>	<u>P_w</u>	<u>P_{node}</u>	Ref.
	(km)		(deg)	(deg)	(deg)	(deg)	(deg/day)	(days)	(yr)	(yr)	
Ariel	190900.	0.0012	115.349	39.481	0.041	22.394	142.8356579	2.520	28.788	57.248	10
Umbriel	266000.	0.0039	84.709	12.469	0.128	33.485	86.8688879	4.144	63.146	126.951	10
Titania	436300.	0.0011	284.400	24.614	0.079	99.771	41.3514246	8.706	161.525	195.369	10
Oberon	583500.	0.0014	104.400	283.088	0.068	279.771	26.7394888	13.46	161.52	195.37	10
Miranda	129900.	0.0013	68.312	311.330	4.338	326.438	254.6906576	1.413	8.913	17.727	10

Mean equatorial orbital elements

Epoch 1986 Jan. 19.50 TT
Solution: URA039 - Jacobson (1998) AJ 115, pp. 1195-1199

Sat.	<u>a</u>	<u>e</u>	<u>w</u>	<u>M</u>	<u>i</u>	<u>node</u>	<u>n</u>	<u>P</u>	<u>P_w</u>	<u>P_{node}</u>	Ref.
	(km)		(deg)	(deg)	(deg)	(deg)	(deg/day)	(days)	(yr)	(yr)	
Cordelia	49800.	0.0003	136.827	254.805	0.085	38.374	1074.5183162	0.335	0.328	0.657	9
Ophelia	53800.	0.0099	17.761	116.259	0.104	164.048	956.4283330	0.376	0.431	0.862	9
Bianca	59200.	0.0009	8.293	138.486	0.193	93.220	828.3879607	0.435	0.603	1.206	9
Cressida	61800.	0.0004	44.236	233.795	0.006	99.403	776.5824144	0.464	0.701	1.402	9
Desdemona	62700.	0.0001	183.285	184.627	0.113	306.089	760.0555393	0.474	0.737	1.474	9
Juliet	64400.	0.0007	223.819	244.696	0.065	200.155	730.1261349	0.493	0.809	1.619	9
Portia	66100.	0.0001	222.433	218.312	0.059	260.067	701.4864814	0.513	0.888	1.777	9
Rosalind	69900.	0.0001	140.477	136.181	0.279	12.847	644.6304183	0.558	1.081	2.163	9
Belinda	75300.	0.0001	42.406	357.224	0.031	279.337	577.3602885	0.624	1.397	2.796	9
Puck	86000.	0.0001	177.094	245.796	0.319	268.734	472.5445878	0.762	2.224	4.448	9

Mean equatorial orbital elements

Epoch 2004 Aug. 25.50 TT
Solution: Showalter (2006) Science, 311, 973

Sat.	<u>a</u>	<u>e</u>	<u>w</u>	<u>M</u>	<u>i</u>	<u>node</u>	<u>n</u>	<u>P</u>	<u>P_w</u>	<u>P_{node}</u>	Ref.
	(km)		(deg)	(deg)	(deg)	(deg)	(deg/day)	(days)	(yr)	(yr)	
Perdita	76417.	0.0116	253.925	192.405	0.470	309.376	564.2469590	0.638	1.479	2.958	51
Mab	97736.	0.0025	249.565	273.769	0.134	350.737	390.0501070	0.923	3.502	7.004	51
Cupid	74392.	0.0013	247.608	163.830	0.099	182.793	587.4436490	0.613	1.346	2.693	51

Mean orbital elements referred to the [Ecliptic](#)

Epoch 2000 Jan. 1.50 TT

Solution: URA090

Sat.	<i>a</i>	<i>e</i>	<i>w</i>	<i>M</i>	<i>i</i>	<i>node</i>	<i>n</i>	<i>P</i>	<i>P_w</i>	<i>P_{node}</i>	Ref.
	(<i>km</i>)		(<i>deg</i>)	(<i>deg</i>)	(<i>deg</i>)	(<i>deg</i>)	(<i>deg/day</i>)	(<i>days</i>)	(<i>yr</i>)	(<i>yr</i>)	
Caliban	7231100.	0.1812	354.339	7.271	141.529	171.189	0.6209758	579.73	8936.31	6445.52	16
Sycorax	12179400.	0.5219	20.103	266.583	159.420	263.034	0.2794217	1288.38	1393.43	1863.29	16
Prospero	16276800.	0.4445	174.152	233.586	151.830	319.003	0.1819679	1978.37	1155.47	1370.39	16
Setebos	17420400.	0.5908	359.953	179.449	158.235	250.235	0.1617921	2225.08	823.91	1044.20	16
Stephano	8007400.	0.2248	14.956	270.163	143.819	191.411	0.5313866	677.47	5484.23	5182.20	16
Trinculo	8505200.	0.2194	158.688	180.374	166.971	193.755	0.4803847	749.40	2592.77	4273.04	16
Francisco	4282900.	0.1324	140.644	3.202	147.250	100.738	1.3478663	267.09	12574.36	14080.40	16
Margaret	14146700.	0.6772	90.017	322.187	57.367	7.067	0.2167366	1661.00	12057739.22	987.08	16
Ferdinand	20430000.	0.3993	156.298	26.163	169.793	217.350	0.1290311	2790.03	747.30	1001.70	16

Satellites of Neptune

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Mean orbital elements referred to the local [Laplace planes](#)

Epoch 2000 Jan. 1.50 TT

Sat.	<i>a</i>	<i>e</i>	<i>w</i>	<i>M</i>	<i>i</i>	<i>node</i>	<i>n</i>	<i>P</i>	<i>P_w</i>	<i>P_{node}</i>	<i>R.A.</i>	<i>Dec.</i>	<i>Tilt</i>	Ref.
	(<i>km</i>)		(<i>deg</i>)	(<i>deg</i>)	(<i>deg</i>)	(<i>deg</i>)	(<i>deg/day</i>)	(<i>days</i>)	(<i>yr</i>)	(<i>yr</i>)	(<i>deg</i>)	(<i>deg</i>)	(<i>deg</i>)	
Triton	354759.	0.0000	66.142	352.257	156.865	177.608	61.2572638	5.877	386.371	687.446	299.456	43.414	0.010	54
Nereid	5513818.	0.7507	281.117	216.692	7.090	335.570	0.9996276	360.13	8091.45	9455.73	269.302	69.117	30.011	54

Mean orbital elements referred to the local [Laplace planes](#)

Epoch 2000 Jan. 1.50 TDT

Solution: Voyager, HST, and Astrometric data solution

Sat.	<i>a</i>	<i>e</i>	<i>w</i>	<i>M</i>	<i>i</i>	<i>node</i>	<i>n</i>	<i>P</i>	<i>P_w</i>	<i>P_{node}</i>	<i>R.A.</i>	<i>Dec.</i>	<i>Tilt</i>	Ref.
	(<i>km</i>)		(<i>deg</i>)	(<i>deg</i>)	(<i>deg</i>)	(<i>deg</i>)	(<i>deg/day</i>)	(<i>days</i>)	(<i>yr</i>)	(<i>yr</i>)	(<i>deg</i>)	(<i>deg</i>)	(<i>deg</i>)	
Naiad	48227.	0.0003	2.045	30.035	4.691	42.279	1222.8435847	0.294	0.289	0.575	299.431	42.940	0.448	55
Thalassa	50074.	0.0002	237.065	262.923	0.135	145.980	1155.7559749	0.311	0.327	0.654	299.431	42.939	0.449	55
Despina	52526.	0.0002	176.857	230.812	0.068	77.060	1075.7330674	0.335	0.387	0.774	299.431	42.937	0.451	55
Galatea	61953.	0.0001	343.011	65.999	0.034	37.247	839.6613030	0.429	0.690	1.380	299.430	42.925	0.462	55
Larissa	73548.	0.0014	249.891	166.246	0.205	308.127	649.0540770	0.555	1.257	2.516	299.429	42.897	0.490	55
Proteus	117646.	0.0005	67.968	250.938	0.075	315.131	320.7656245	1.122	6.380	12.727	299.406	42.432	0.974	54
S/2004 N 1	105284.	0.0000	0.000	302.652	0.000	0.000	378.9061402	0.950	0.000	0.000	299.373	42.944	0.448	57

Mean [ecliptic](#) orbital elements

Epoch 2003 Jun. 10.00 TT

Sat.	<i>a</i>	<i>e</i>	<i>w</i>	<i>M</i>	<i>i</i>	<i>node</i>	<i>n</i>	<i>P</i>	<i>P_w</i>	<i>P_{node}</i>	Ref.
	(<i>km</i>)		(<i>deg</i>)	(<i>deg</i>)	(<i>deg</i>)	(<i>deg</i>)	(<i>deg/day</i>)	(<i>days</i>)	(<i>yr</i>)	(<i>yr</i>)	
Halimede	16681000.	0.2909	162.119	105.258	112.898	217.288	0.1915578	1879.33	5673.19	18501.48	56
Psamathe	46705000.	0.4617	144.158	190.027	137.679	298.074	0.0394359	9128.74	1477.62	1304.72	56
Sao	22619000.	0.2827	65.047	168.139	49.907	60.354	0.1233230	2919.16	436880.78	5382.39	56
Laomedeia	23613000.	0.4339	140.107	285.863	34.049	59.124	0.1133637	3175.62	2245.08	3325.44	56

Satellites of Plutojump to: [Earth](#), [Mars](#), [Jupiter](#), [Saturn](#), [Uranus](#), [Neptune](#), [Pluto](#) or [\[top of page\]](#)

Mean equatorial (IAU equator) orbital elements, planetocentric for Charon, barycentric for the others

Solution: PLU042

Sat.	<u>a</u>	<u>e</u>	<u>w</u>	<u>M</u>	<u>i</u>	<u>node</u>	<u>n</u>	<u>P</u>	<u>P_w</u>	<u>P_{node}</u>	Ref.
	(km)		(deg)	(deg)	(deg)	(deg)	(deg/day)	(days)	(yr)	(yr)	
Charon	19591.	0.0002	146.106	131.070	0.080	26.928	56.3625210	6.387	10178.040	9020.398	58
Nix	48671.	0.0000	324.463	284.405	0.000	203.400	14.4841304	24.85	0.000	0.000	58
Hydra	64698.	0.0056	153.307	326.678	0.304	113.173	9.4235386	38.20	7.19	14.35	58
Kerberos	57729.	0.0000	160.629	161.061	0.426	305.871	11.1913026	32.17	0.000	9.27	58
Styx	42393.	0.0006	330.244	194.546	0.080	26.956	17.8551357	20.16	3710.58	5162.64	58

Common Table Column Headings:

- a* Semi-major Axis (mean value)
- e* Eccentricity (mean value)
- w* Argument of periapsis (mean value)
- M* Mean anomaly (mean value)
- i* Inclination with respect to the reference plane: [ecliptic](#), ICRF, or [local Laplace](#) (mean value)
- node* Longitude of the ascending node (mean value) measured from the node of the reference plane on the ICRF equator
- n* Longitude rate (mean value)
- P* Sidereal period (mean value)
- P_w* Argument of periapsis precession period (mean value)
- P_{node}* Longitude of the ascending node precession period (mean value)

Headings only for elements with respect to the local [Laplace plane](#):

- R.A.* Right ascension and ...
- Dec.* Declination of the [Laplace plane](#) pole with respect to the ICRF.
- Tilt* The angle between the planet equator and the [Laplace plane](#).

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